

```

import pandas as pd
df=pd.read_excel("Data_Cleaning_50Plus_Records.xlsx")
df.head(10)
print(df.columns)

Index(['Student ID', 'Student Name ', ' Department', 'Age', 'Salary',
      'Email'], dtype='object')

print(df.columns)

Index(['Student ID', 'Student Name ', ' Department', 'Age', 'Salary',
      'Email'], dtype='object')

df.rename(columns={'Student ID':'Student_ID','Student
Name':'Student_Name','Department':'Department'}, inplace=True)

print(df.columns)

Index(['Student_ID', 'Student Name ', ' Department', 'Age', 'Salary',
      'Email'], dtype='object')

print(df.dtypes)

Student_ID          int64
Student Name        object
Department           object
Age                 float64
Salary              float64
Email               object
dtype: object

print(df.duplicated().sum())

0

print(df.shape)

(52, 6)

df.drop_duplicates(inplace=True)

print(df.shape)

(52, 6)

import pandas as pd
df=pd.read_excel("Data_Cleaning_50Plus_Records.xlsx")
import pandas as pd
df=pd.read_excel("Data_Cleaning_50Plus_Records.xlsx")
df['Student Name '] = df['Student Name '].str.strip()
df['Student Name '] = df['Student Name '].str.upper()

```

```
print(df['Student Name '].head())
df['Student Name '] = df['Student Name '].str.title()
print(df['Student Name '].head())
```

```
0    STUDENT1
1    STUDENT2
2    STUDENT3
3    STUDENT4
4    STUDENT5
Name: Student Name , dtype: object
```

```
0    Student1
1    Student2
2    Student3
3    Student4
4    Student5
```

```
Name: Student Name , dtype: object
```

```
df[' Department'] = df[' Department'].replace({
    'CSE':'Computer Science',
    'ECE':'Eletronics and communi eng',
    'EEE':'electrical and electronic',
    'IT':'Information tech',
    'MECH':'Mechanical'})
```

```
print(df[' Department'].head())
```

```
0    Information tech
1    Eletronics and communi eng
2    electrical and electronic
3    Mechanical
4    Computer Science
```

```
Name: Department, dtype: object
```

```
df['Name Length'] = df['Student Name '].str.len()
```

```
print(df[['Student Name ', 'Name Length']].head())
```

```
print(df.columns)
```

```
Student Name  Name Length
0    Student1           8
1    Student2           8
2    Student3           8
3    Student4           8
4    Student5           8
```

```
Index(['Student ID', 'Student Name ', ' Department', 'Age', 'Salary',
      'Email',
      'Name Length'],
      dtype='object')
```

```
df['First_Letter'] = df['Student Name '].str[0]  
print(df[['Student Name ', 'First_Letter']].head())
```

	Student Name	First_Letter
0	Student1	S
1	Student2	S
2	Student3	S
3	Student4	S
4	Student5	S

```
!pip install matplotlib
```